MAME 0.289 on the Raspberry Pi 5

Optimization journal: compiler, driver selection, PGO and runtime settings

Raspberry Pi 5 (8 GB, arm_freq 3000 MHz), RetroPie on Debian 13 Trixie, GCC 14.2, Mesa / V3DV. Work done from 9 to 11 November 2026.

1. Summary

The MAME standalone binary installed by the *mamedev.sh* scriptmodule is a generic AArch64 build. Rebuilding MAME on the Pi itself, for the Cortex-A76, with only the drivers of the collection, link-time optimization (LTO) and profile-guided optimization (PGO) raised the speed of the reference 3D game by **one third**. The display back-end (accel, OpenGL, bgfx) does not change the speed: the emulated CPU and the software 3D renderer are the limit.

Build / setting	Samurai Shodown 64, in-fight	vs prebuilt
Prebuilt generic binary (GCC 12.2, Bookworm)	58.6 %	-
Full MAME, -mcpu=cortex-a76, GCC 14, -O3	65.5 %	+12 %
Collection drivers only + LTO	67.2 % (67.4 / 67.6 in other runs)	+15 %
Collection drivers + LTO + PGO (installed)	**77.6 - 78.8 %**	**+33 %**
+ advanced LTO flags (-fipa-pta ...)	77.3 %	no gain, rejected
Runtime: numprocessors 2, governor performance in games	best measured values	kept

Slow drivers gained as much or more: 39in1 +47 %, gp32 +36 %, ssfindo +38 % (now above 100 %), bballoon +32 %, turrett +20 %. Final binary: 196 MB (prebuilt: 444 MB). Free disk space at the end: 25 GB (26 GB before the work).

2. Measurement method

All speed figures come from MAME's own *Average speed* line, with the CPU cooled below 55 °C before each run, runs interleaved (A B B A A B) and *vcgencmd get_throttled* = 0x0 at the end. Measurements taken while the Pi was on a weak supply (under-voltage 0x50000) were discarded and redone.

Item	Value
Reference game	Samurai Shodown 64 (sams64, Hyper Neo Geo 64): MIPS R4300 through the ARM64 DRC plus software 3D
Scene	a mid-fight save state (quick.sta) loaded 5 s after boot: -autoboot_delay 5 -autoboot_script load_quick.lua (manager.machine:load("quick")). -state alone loaded too early and was ignored.
Command	-str 60 -nothrottle -noautoframeskip -frameskip 0 -video none -sound none -numprocessors 2
Driver classification	-bench 8 (8 emulated seconds, boot / attract) for one game per driver
Display tests	same scene launched from EmulationStation (X sessions cannot be started from SSH)

3. Starting point: the prebuilt binary

The installed binary carried the signature *GCC: (Debian 12.2.0-14+deb12u1)* (Bookworm) while the Pi runs Trixie with GCC 14.2. Its *retropie.pkg* said *pkg_origin="source"* because it was built with RetroPie-Setup on the packager's machine (time zone +02:00).

Generic code detected with atomic instructions. A Cortex-A76 build inlines ARMv8.1 LSE atomics (cas, ldadd, swp); a generic armv8-a build calls the *__aarch64_** out-of-line helpers instead:

```
objdump -d --no-show-raw-insn mame | awk -F'\t' '$2~/^(casa?l?|ldadda?l?|swpa?l?)$/{l++} /__aarch64_/{g++} END{print
-> LSE: 11 (inside the helpers)   generic helper calls: 2080
```

First settings checks with this binary (attract mode, display off):

Test	Result
numprocessors 1 / 2 / auto / 3 / 4	76.0 / 75.1 / 75.0 / 73.0 / 70.9 %
ARM64 DRC on / off	76.6 % / 55 %
-multithreading	option no longer exists in 0.289 ("unknown option")
Duplicate 'video' lines in mame.ini	the last one wins: 'soft' was silently in use; cleaned
In-fight, display on: accel / opengl	57.4 % / 57.5 % (2 runs each)

4. bgfx: crash found and fixed

bgfx crashed with both back-ends right after *Initializing BGFX library*. Two separate causes:

Back-end	Cause	Fix
GLES	the test line ran outside X: bgfx needs an X11 window handle	launch with XINIT-WM: (works)
Vulkan	gdb trace (debug build): SIGSEGV in libvulkan_broadcom.so from bgfx::vk::TextureVK::createView <- SwapChainVK::createAttachments. bgfx maps D24S8 to VK_FORMAT_D32_SFLOAT_S8_UINT, which V3DV does not support; the image is created anyway and the driver crashes on the view. The same table is still in upstream bgfx.	renderer_vk.cpp, table line '// D24S8': D32_SFLOAT_S8_UINT -> D24_UNORM_S8_UINT. bgfx already checks support and falls back if needed. Vulkan then works.

```
sed -i -E '/\/\/ D24S8\s*$/ s/VK_FORMAT_D32_SFLOAT_S8_UINT/VK_FORMAT_D24_UNORM_S8_UINT/' \
    3rdparty/bgfx/src/renderer_vk.cpp
```

Display back-ends with the same binary (hng64 build, -O3, no LTO), in-fight, sound on, launched from EmulationStation:

accel (no X)	bgfx Vulkan (X)	bgfx GLES (X)	OpenGL (X)
68.5 %	69.1 %	69.0 %	68.0 %

All within one point (noise). bgfx already renders on its own thread; MAME draws the emulated 3D in software, so the GPU cannot take that work. bgfx is kept for visual filters only (CRT, LCD), at no cost.

5. Compiler tests

Test	Runs (sams64 in-fight)	Mean	Gain
Prebuilt (generic)	56.32 / 58.83 / 60.26	58.5 %	ref
hng64 driver only, -mcpu=cortex-a76 -O3 LTO, GCC 14	72.21 / 71.34 / 72.87	72.1 %	+23 %
Prebuilt (second session)	60.61 / 60.03 / 56.14	58.9 %	ref
Full MAME, cortex-a76, -O3, GCC 14, no LTO (LTO needs 16-32 GB)	65.55 / 65.71 / 66.13	65.8 %	+12 %
Full MAME (same session as below)	66.19 / 65.64 / 64.64	65.5 %	-
Collection drivers only (980) + LTO	67.24 / 67.02 / 68.56	67.6 %	+3 % vs full
Collection + LTO (session of the PGO test)	67.90 / 66.32 / 67.25	67.2 %	ref
Collection + LTO + PGO	78.47 / 79.04 / 78.64	78.7 %	+17 % vs LTO
PGO (session of the advanced-flags test)	77.75 / 77.89 / 77.26	77.6 %	ref
PGO + -fipa-pta -fipa-icf -fdevirtualize-at-ltrans -fgcse-sm -fgcse-las -fvariable-expansion-in-unroller	77.76 / 77.67 / 76.42	77.3 %	none

The single-driver hng64 binary was fast mostly because it was tiny (the hot code fits in the caches). With 980 drivers (183 MB) that advantage shrinks to +3 %; PGO, which groups hot code and optimizes the branches actually taken, recovered and exceeded it. The advanced LTO flags quoted by MAME's developers (~2 %) bring nothing once PGO is used.

Five slowest drivers (excluding non-working 'skeleton' machines), 20 emulated seconds, one run each:

Driver	LTO	PGO	Gain	PGO + advanced flags
39in1 (misc/39in1, ARM)	21.4 %	31.4 %	+47 %	31.3 %
bballoon (eolith/ghosteo, ARM9)	66.9 %	88.6 %	+32 %	87.8 %
turrett (namco/turrett)	123.4 %	148.1 %	+20 %	145.1 %
gp32 (gamepark/gp32, ARM)	68.3 %	92.6 %	+36 %	87.5 %
ssfindo (acorn/ssfindo)	81.4 %	112.0 %	+38 %	106.0 %

6. Building only the collection's drivers

MAME can be built with *SOURCES*= (a comma list of driver files). The list was produced from the ROM folders and from every *emulators.cfg*:

Folder type (from its emulators.cfg)	Drivers taken
MAME line without machine name (arcade-style: mame %BASENAME%)	every .zip / .7z file name and CHD folder name that is a MAME machine (all_in1, classich, jakks, konamih, tigerh, arcade ...)
MAME line with a machine name (mame supracan -cart %ROM%)	that machine only: the files are software-list items
Both kinds of lines	both (e.g. playchoice10, supracan, amiga, jaguar)
No MAME standalone line	ignored (consoles run by other emulators; names matching by chance)
Forced folders (dossiers_forces.txt)	treated as arcade-style even without a MAME line
Excluded folders (systemes_choisis.txt)	13 removed by hand: amigacd32, atarijaguar, atomiswave, coleco_sgm, jaguar, jaguarcd, neogeo, neogeo-cd, naomi, naomi2, pcecd, sega_system32, tg-cd

Count	Value
Driver source files in MAME 0.289 (mame.lst)	3,349 (7,274 entries with -listsource, devices included)
Games and machines kept	3,933
Driver files needed	980 (4 'devices/' entries skipped: compiled automatically)
With companion files (video, sound, 3D)	1,340, then 1,341 (xavix_2000.cpp)
Binary size	183 MB with LTO, 196 MB with PGO (full MAME: 375 MB)

Companion files: MAME's makedep only adds *name_a/_v/_m.cpp*; files such as *hng64_3d.cpp* are added when they share the driver's prefix and contain no machine definition (GAME/CONS/COMP/SYST). One cross-file dependency appeared only in the

non-LTO (instrumented) link: *xavix_m.cpp* needs *xavix_duelmast_state*, defined in *xavix_2000.cpp*. With LTO the unused reference was dropped.

7. Driver classification

One game per driver, 8 emulated seconds (boot / attract), with the LTO build: 974 drivers measured in about 40 minutes.

Heavy (<100 %)	Medium (100-200 %)	Light (>200 %)	Did not start
7	14	778	175 (missing BIOS / files, media needed)

Slowest: zeebo 20 % (skeleton), 39in1 22 %, bballoon 38 %, digiblst 47 % (skeleton), turrett 64 %, gp32 68 %, ssfindo 85 %, bbust2 109 %. Boot-time figures under-rate 3D systems: sams64 is 67 % in a fight but its driver showed 109 % at boot, so PGO training used a wide threshold (400 %).

8. Profile-guided optimization

Step	Details
Instrumented build	-fprofile-generate=/home/pi/mame_build/pgo -fprofile-update=prefer-atomic, no LTO, -j3, 8 GB extra swap. The instrumented binary runs 4-5x slower (sams64 fight: 15 %).
Training	sams64: 120 s of the fight state + 90 s of attract mode; then one game per driver under 400 % (54 drivers, skeleton excluded): 15 emulated s under 100 %, 30 s otherwise. timeout -s INT 1200 so MAME exits cleanly and writes its profile. 56 sessions, 0 failures, 3,996 profile files, about 1.5 h.
Final build	-fprofile-use -fprofile-partial-training -fprofile-correction -Wno-missing-profile -Wno-coverage-mismatch, LTO=1, -Wl,--no-keep-memory, -j3. Untrained code (bgfx, light drivers) stays normally optimized thanks to -fprofile-partial-training.
Check	every one of the 3,933 names present in the new binary (-listsource); played tests on sams64 (state), arcade, all_in1, classich, jakks, supracan

9. Runtime settings

Setting	Measured	Decision
numprocessors 1 / 2 / 3 / 4 (PGO, sound off)	77.9 / **78.8** / 77.6 / 76.0 %	keep 2
Sound on vs off	77.9 % vs 78.5 %: ~0.6 point	no change (no resampler option in 0.289; modules: sdl, portaudio, pulse, pipewire, none)
CPU governor during a game	ondemand: 1800-3000 MHz, 5 of 15 samples below max; performance: 3000 MHz steady, same temperature (50-52 °C)	governor = "performance" in runcommand.cfg (only while a game runs; restored afterwards)
Display back-end	all within 1 point (section 4)	accel by default; bgfx Vulkan line for filters

10. Pitfalls met and how they are handled

Problem	Symptom	Fix
Build tree with a space in its path	MAME's make cannot handle 'arcade temp'	build in /home/pi/mame_build
libpipewire-0.3-dev	apt conflict: PipeWire comes from trixie-backports	apt install -t trixie-backports libpipewire-0.3-dev
XINIT from SSH	'vt/' then 'Permission denied' on the VT; root X locked the Pi	launch X tests from EmulationStation only
Default emulator pointing to a removed line	'Executing:' empty in runcommand.log	reset default; delete per-ROM keys in all/emulators.cfg
Generated makefiles not regenerated	new SOURCES or ARCHOPTS silently ignored	start from an empty build/ (or delete build/projects/sdl/mamemonmame)
Instrumented build out of memory	4 compilers in state D, 0 % CPU, 2 GB swap full	-j3 and an 8 GB swap file
Non-LTO link	undefined reference to xavix_duelmast_state	add the defining file (xavix_2000.cpp)
Profile mismatch	'does not match its profile data' (rapidjson, m6809) treated as an error	-Wno-coverage-mismatch
Training run killed by timeout	no .gcda written for that driver	timeout -s INT and shorter runs for slow drivers
LTO whole-program analysis	swap peaked at 8.9 GB with -fipa-pta	8 GB extra swap file during LTO
Heat during LTO	throttled=0x80000 (soft limit 80 °C), fan already at 100 %	harmless, only slower
Long pasted lines in PuTTY	scripts cut in the middle	short lines, scripts written in numbered chunks with an end marker
mamedev package update	would reinstall the generic prebuilt binary	do not update mamedev; rebuild with the script instead

11. Build times and resources

Build	Time	Notes
hng64 only, LTO / -O3 with symbols	~45 min / ~40 min	test builds
Full MAME, -O3, no LTO	~3.5 h	375 MB
Collection + LTO	2 h 42	LTO peak handled by 8 GB extra swap
Collection, instrumented	~2.5 h (+ one full rebuild after the regeneration issue)	-j3
Collection, PGO + LTO	~3 h	196 MB
Training sessions	~1.5 h	56 sessions
Disk	26 GB free before, 25 GB after	PGO profile archive: 11 MB; source tree removable (4.9 GB)

12. Final configuration

Item	Value
Binary	/opt/retropie/emulators/mame/mame, 0.289 (mame0289-dirty), GCC 14.2, 196 MB, PGO + LTO
mame.ini	video accel, numprocessors 2, autoframeskip 1, samplerate 44100 (48000 optional, quality only)
runcommand.cfg	governor = "performance"
Fallback	monmame_lto (LTO build, 183 MB) in the data folder
Kept for updates	scripts, monmame_sources.txt, systemes_choisis.txt, dossiers_forces.txt, classement_pilotes.txt, pgo_mesures.tar.gz, renderer_vk.cpp (fixed and original)

emulators.cfg lines. Every MAME system keeps its default line. A *mame_bgfx_vulkan* line is added, built from the system's default MAME line with the same arguments, so that bgfx can be chosen per game in the runcommand menu. Obsolete lines (test lines, missing binaries, machines not in this MAME, options rejected by MAME) are removed after

a report. Examples:

```
# /opt/retropie/configs/hyperneogeo64/emulators.cfg
mame = "/opt/retropie/emulators/mame/mame %BASENAME%"
mame_bgfx_vulkan = "XINIT-WM:VK_INSTANCE_LAYERS= VK_LAYER_PATH= MESA_VK_WSI_PRESENT_MODE=relaxed
   /opt/retropie/emulators/mame/mame -video bgfx -bgfx_backend vulkan
   -bgfx_path /opt/retropie/emulators/mame/bgfx -bgfx_screen_chains default -waitvsync 0 %BASENAME%"
default = "mame"

# /opt/retropie/configs/supracan/emulators.cfg (machine-style system)
mame_bgfx_vulkan = "XINIT-WM:... /opt/retropie/emulators/mame/mame -video bgfx ...
   -v -c supracan -cart %ROM%"

# /opt/retropie/configs/all/runcommand.cfg
governor = "performance"
```

(The bgfx lines are written on one line in the real files; they are wrapped here for printing. The bgfx options live in bgfx_options.txt so that every line can be regenerated after tuning.)

13. Reproducing it, or updating to MAME 0.290 and later

mame_pi5_pgo_build.sh does the whole procedure without the investigation tests. Phases:

```
deps      build dependencies (PipeWire -dev from trixie-backports)
fetch     git clone --depth 1 --branch $MAME_TAG + bgfx Vulkan fix
detect    driver list: ROM folders + emulators.cfg + mame.lst + companions
classify  -bench 8 per driver with the installed MAME (~40 min)
gen       instrumented build (adds files for undefined references)  ~2.5-3 h
train     state fight + attract + one game per driver < 400 %        ~1-1.5 h
use       PGO + LTO build                                            ~3-3.5 h
verify    every detected game present in the new binary
install   install (previous binary kept as mame.previous)
tune      numprocessors 2 + governor performance
cfg-report / cfg-apply   emulators.cfg cleanup + mame_bgfx_vulkan lines
clean     archive the profile, remove swap file and build files

Full rebuild (one night):
  MAME_TAG=mame0290 nohup bash mame_pi5_pgo_build.sh full > full.log 2>&1 < /dev/null &
  bash mame_pi5_pgo_build.sh install tune cfg-report
  bash mame_pi5_pgo_build.sh cfg-apply          (after reading the report)

Fast update (~3.5 h), reusing the 0.289 profile (unchanged functions keep it):
  MAME_TAG=mame0290 nohup bash mame_pi5_pgo_build.sh fast > fast.log 2>&1 < /dev/null &
```

Check before updating: romsets renamed between versions (they show up as unrecognized names), drivers moved in the source tree (handled by mame.lst), and whether the bgfx fix is still needed (the fetch phase reports it). Updating every month is not necessary.

14. Tried and rejected

Attempt	Result
bgfx (Vulkan or GLES) for speed	same speed as accel; kept only for filters
Advanced LTO flags on top of PGO	77.3 % vs 77.6 %: noise; extra 1.5 h and 8.9 GB swap
LTO on full MAME	needs 16-32 GB of RAM; Linux distributions disabled it too (openSUSE)
Cross-compiling on the PC	8 GB of RAM (less than the Pi has free); no benefit
numprocessors 3 or 4	slower (-1 and -3 points)
Sound tuning	sound costs ~0.6 point only
Zink (OpenGL over Vulkan)	V3DV lacks nullDescriptor (robustness2): refused
Overclock above 3 GHz, isolcpus, Clang	not tried: heat limit already reached, negligible or uncertain gain